# GEOMETRIC REDUCEDNESS OF THE GENERIC FIBRE SPREADS TO ALL FIBRES OVER A DENSE OPEN

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 7214-7225 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 7227-7256, the complete author proof (28 body lines).

*Editorial note.* Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Geometric reducedness of fibres, base change to the algebraic closure, and finite presentation of the relevant algebras; the spreading argument is retained.

*Source reference.* Original lemma lemma-scheme-theoretically-dense-generic-fibre: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6563–6570. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 056D: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7256–7256. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 056E: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7256–7256. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02V4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7256–7256. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 052A: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7256–7256. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01ON: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7256–7256. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 056V: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7256–7256. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : X \to Y$ be a morphism of schemes. Assume*

(1) *$Y$ is irreducible with generic point $\eta$,*
(2) *$X_\eta$ is geometrically reduced, and*
(3) *$f$ is of finite type.*

*Then there exists a nonempty open subscheme $V \subset Y$ such that $X_V \to V$ has geometrically reduced fibres.*

*Proof.* Let $Y' \subset Y$ be the reduction of $Y$. Let $X' \to Y'$ be the base change of $f$. Note that $Y' \to Y$ induces a bijection on points and that $X' \to X$ identifies fibres. Hence we may assume that $Y'$ is reduced, i.e., integral, see Properties, Lemma 01ON. We may also replace $Y$ by an affine open. Hence we may assume that $Y = \mathrm{Spec}(A)$ with $A$ a domain. Denote $K$ the fraction field of $A$. After shrinking $Y$ a bit we may also assume that $X \to Y$ is flat and of finite presentation, see Morphisms, Proposition 052A.

As $X_\eta$ is geometrically reduced there exists an open dense subset $V \subset X_\eta$ such that $V \to \mathrm{Spec}(K)$ is smooth, see Varieties, Lemma 056V. Let $U \subset X$ be the set of points where $f$ is smooth. By Morphisms, Lemma 02V4 we see that $V \subset U_\eta$. Thus the generic fibre of $U$ is dense in the generic fibre of $X$. Since $X_\eta$ is reduced, it follows that $U_\eta$ is scheme theoretically dense in $X_\eta$, see Morphisms, Lemma 056D. We note that as $U \to Y$ is smooth all the fibres of $U \to Y$ are geometrically reduced. Thus it suffices to show that, after shrinking $Y$, for all $y \in Y$ the scheme $U_y$ is scheme theoretically dense in $X_y$, see Morphisms, Lemma 056E. This follows from Lemma lemma-scheme-theoretically-dense-generic-fibre. □